

three generations of matter
(fermions)

interactions / force carriers
(bosons)

I

II

III

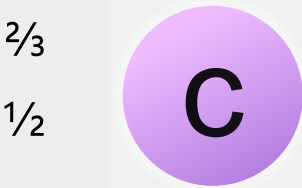
mass
charge
spin

$\approx 2.16 \text{ MeV}/c^2$



up

$\approx 1.273 \text{ GeV}/c^2$



charm

$\approx 172.57 \text{ GeV}/c^2$



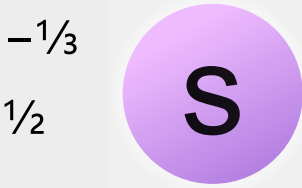
top

$\approx 4.7 \text{ MeV}/c^2$



down

$\approx 93.5 \text{ MeV}/c^2$



strange

$\approx 4.183 \text{ GeV}/c^2$



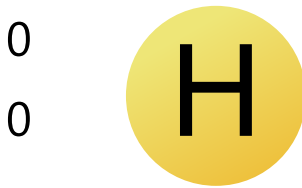
bottom

0



gluon

$\approx 125.2 \text{ GeV}/c^2$



higgs

$-\frac{1}{3}$

$\frac{1}{2}$

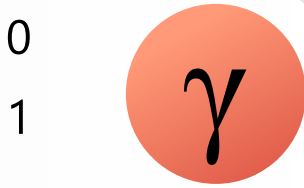
$-\frac{1}{3}$

$\frac{1}{2}$

$-\frac{1}{3}$

$\frac{1}{2}$

0



photon

0

1

$\approx 0.511 \text{ MeV}/c^2$

-1

$\frac{1}{2}$



electron

$\approx 105.66 \text{ MeV}/c^2$

-1

$\frac{1}{2}$



muon

$\approx 1.77693 \text{ GeV}/c^2$

-1

$\frac{1}{2}$



tau

$\approx 91.188 \text{ GeV}/c^2$

0

1



Z boson

$< 0.8 \text{ eV}/c^2$

0

$\frac{1}{2}$



**electron
neutrino**

$< 0.17 \text{ MeV}/c^2$

0

$\frac{1}{2}$



**muon
neutrino**

$< 18.2 \text{ MeV}/c^2$

0

$\frac{1}{2}$



**tau
neutrino**

$\approx 80.3692 \text{ GeV}/c^2$

± 1

1



W boson

GAUGE BOSONS
VECTOR BOSONS

SCALAR BOSONS

QUARKS

LEPTONS